

# AI ENGINEER

Complete Study Roadmap

Tailored for the Senpage Consulting Internship

Includes: Core AI · ML · Deep Learning · LLM Engineering · NLP · Fine-Tuning · Backend · DevOps

**10**

Learning Phases

**5+**

Portfolio Projects

**12+**

Technologies

**6-12**

Months Plan

## Topics

- OOP & Design Patterns
- Decorators & Metaclasses
- Generators & Iterators
- Async/Await & Concurrency
- Context Managers
- Type Hints & Pydantic
- Packaging & Virtual Envs
- Testing with pytest

## Resources

- Real Python (realpython.com)
- Python Asyncio docs
- Fluent Python (book)
- pytest official docs
- Build a mini LangChain clone
- Build a Python package & publish

## Topics

- Arrays, Strings, Hashing
- Linked Lists, Stacks, Queues
- Trees (BST, AVL, Tries)
- Graphs (BFS/DFS/Dijkstra)
- Heaps & Priority Queues
- Dynamic Programming
- Binary Search
- Greedy Algorithms

## Resources

- LeetCode Easy → Medium (150–200)
- NeetCode 150 roadmap
- AlgoExpert
- CLRS book (reference)
- LeetCode Blind 75 list
- Daily 2–3 problems habit

| Topics                                   | Resources                          |
|------------------------------------------|------------------------------------|
| • Linear Algebra (vectors, matrices)     | • fast.ai Practical ML             |
| • Statistics & Probability               | • StatQuest (YouTube)              |
| • Supervised: Regression, Classification | • Hands-On ML (Aurélien Géron)     |
| • Unsupervised: Clustering, PCA          | • Kaggle Learn free courses        |
| • Model Evaluation & Cross-Validation    | • Projects: House Price Prediction |
| • Feature Engineering                    | • Customer Churn Prediction        |
| • NumPy, Pandas, Scikit-learn            | • Recommendation System            |
| • Matplotlib & Seaborn                   |                                    |

Phase 4

## Deep Learning with PyTorch

■ 4–8 We

| Topics                                | Resources                                |
|---------------------------------------|------------------------------------------|
| • Neural Networks & Backprop          | • fast.ai Deep Learning                  |
| • CNNs (image tasks)                  | • d2l.ai (Dive into DL)                  |
| • RNNs & LSTMs (sequential data)      | • PyTorch official tutorials             |
| • Transformers & Attention            | • Andrej Karpathy (YouTube)              |
| • Batch Norm, Dropout, Regularization | • Projects: Image Classification         |
| • Loss Functions & Optimizers         | • Sentiment Analysis                     |
| • Transfer Learning                   | • Build a small Transformer from scratch |
| • PyTorch Lightning                   |                                          |

## Core NLP Concepts

### Topics

- Tokenization & Stemming/Lemmatization
- Bag-of-Words, TF-IDF
- Word Embeddings (Word2Vec, GloVe, FastText)
- Named Entity Recognition (NER)
- POS Tagging & Dependency Parsing
- Text Classification
- Question Answering
- Summarization & Translation

### Resources

- spaCy docs & tutorials
- NLTK book (free online)
- HuggingFace NLP Course
- Stanford NLP CS224N lectures
- Projects: NER on custom dataset
- Text Summarizer
- Intent Classifier

## Transformer-Based NLP

### Topics

- BERT, RoBERTa, DistilBERT
- GPT architecture family
- T5 & BART (seq2seq)
- Sentence Transformers / SBERT
- Cross-Encoders vs Bi-Encoders
- Embeddings & Semantic Search
- Zero-shot & Few-shot Classification
- NLP Evaluation Metrics (BLEU, ROUGE, F1)

### Resources

- HuggingFace Transformers library
- SBERT documentation
- Papers: Attention Is All You Need
- Papers: BERT (Devlin et al.)
- Projects: Semantic Search Engine
- Q&A; Bot with BERT
- Multi-label Text Classifier

| Topics                                   | Resources                            |
|------------------------------------------|--------------------------------------|
| • Prompt Engineering & Templates         | • LangChain docs                     |
| • RAG (Retrieval Augmented Generation)   | • LlamaIndex docs                    |
| • Chunking Strategies                    | • RAG from Scratch (DeepLearning.AI) |
| • Embeddings & Vector Similarity         | • Qdrant documentation               |
| • Vector DBs: Qdrant, Weaviate, Pinecone | • Projects: AI Tutor Platform        |
| • Retrieval Strategies (MMR, HyDE)       | • Research Agent (multi-step)        |
| • LangChain & LangGraph                  | • Customer Support Bot               |
| • LlamaIndex & CrewAI                    |                                      |

Phase 7

## Fine-Tuning LLMs

■ 3–5 Weeks

### Fine-Tuning Techniques

| Topics                                 | Resources                                   |
|----------------------------------------|---------------------------------------------|
| • Full Fine-Tuning vs PEFT             | • HuggingFace PEFT library                  |
| • LoRA (Low-Rank Adaptation)           | • QLoRA paper (Dettmers et al.)             |
| • QLoRA (Quantized LoRA)               | • TRL library (SFT Trainer)                 |
| • PEFT with HuggingFace                | • Axolotl fine-tuning framework             |
| • Supervised Fine-Tuning (SFT)         | • Unsloth (fast LoRA training)              |
| • Instruction Tuning                   | • Projects: Fine-tune Llama3 on custom Q&A; |
| • RLHF overview                        | • Domain-Specific Classifier via SFT        |
| • DPO (Direct Preference Optimization) |                                             |

### Evaluation & Data Preparation

| Topics                                            | Resources                                  |
|---------------------------------------------------|--------------------------------------------|
| • Dataset Curation & Formatting (Alpaca/ShareGPT) | • Datasets library (HuggingFace)           |
| • Data Cleaning & Deduplication                   | • LM Evaluation Harness                    |
| • GGUF / GGML Quantization                        | • llama.cpp for local serving              |
| • Merging Adapters & Exporting                    | • vLLM for production serving              |
| • Eval: Perplexity, MMLU, HellaSwag               | • Weights & Biases (experiment tracking)   |
| • Serving fine-tuned models (Ollama, vLLM)        | • Projects: Deploy fine-tuned model as API |



## Topics

- RAG Architecture Patterns
- Multi-Agent System Design
- Event-Driven AI Pipelines
- Microservices for AI
- API Design Best Practices
- Caching (Redis)
- Message Queues (RabbitMQ)
- Observability & Monitoring

## Resources

- System Design Interview (Alex Xu)
- Designing ML Systems (Chip Huyen)
- Redis documentation
- RabbitMQ tutorials
- Projects: Distributed Agent Platform
- AI Workflow Engine
- Multi-Agent Orchestration System

## Topics

- FastAPI fundamentals & routing
- REST API design
- Authentication (JWT, OAuth2)
- Middleware & Dependency Injection
- Background tasks & WebSockets
- PostgreSQL + SQLAlchemy ORM
- Alembic migrations
- Testing APIs with pytest

## Resources

- FastAPI official docs
- SQLAlchemy docs
- TestDriven.io FastAPI course
- PostgreSQL tutorial
- Projects: AI SaaS Backend
- Async Task Queue (Celery + Redis)
- Full CRUD API with auth

| Topics                                   | Resources                                    |
|------------------------------------------|----------------------------------------------|
| • Docker & docker-compose                | • Docker official docs                       |
| • GitHub Actions CI/CD                   | • GitHub Actions docs                        |
| • Linux & Bash scripting                 | • The Odin Project (Linux)                   |
| • Nginx reverse proxy                    | • Next.js tutorial                           |
| • Environment management (.env, secrets) | • Projects: Dockerized FastAPI + React App   |
| • React fundamentals                     | • CI/CD pipeline with GitHub Actions         |
| • Next.js basics                         | • Deploy RAG App to cloud (Railway / Render) |
| • Tailwind CSS                           |                                              |



# Portfolio Projects

1

## AI Tutor Platform

- FastAPI + PostgreSQL backend
- LangGraph for multi-step tutoring flow
- RAG over course materials
- React + Tailwind frontend
- Docker deployment

*Demonstrates: full-stack AI, RAG, LangGraph, backend engineering*

2

## Multi-Agent Research Assistant

- CrewAI / LangGraph agents
- Tavily / Serper for web search
- Auto-generates research reports
- Supports PDF export
- LlamaIndex for doc indexing

*Demonstrates: agent systems, tool use, LLM orchestration*

3

## RAG Knowledge Base

- LlamaIndex + Qdrant vector DB
- OpenAI / local LLM (Ollama)
- Document upload (PDF, DOCX)
- Semantic search + chat interface
- Evaluation with RAGAs framework

*Demonstrates: RAG architecture, vector search, evaluation*

4

## NLP Pipeline + Fine-Tuned Classifier

- Fine-tune DistilBERT via QLoRA
- Custom domain text classification
- spaCy NER pipeline
- HuggingFace deployment (Spaces)
- Evaluation dashboard

*Demonstrates: NLP, fine-tuning, model deployment*

5

## AI Resume Analyzer

- Upload CV (PDF) → parse with LLM
- Score against job descriptions
- Suggest improvements with RAG
- FastAPI + React
- PostgreSQL history storage

*Demonstrates: document AI, RAG, full-stack product thinking*

# Study Timelines

## 3-Month Sprint (Minimum Viable)

| Topic                      | Timeline   |
|----------------------------|------------|
| Python Advanced            | Week 1-2   |
| DSA Basics (arrays→graphs) | Week 2-4   |
| ML Fundamentals            | Week 4-6   |
| PyTorch Deep Learning      | Week 6-9   |
| RAG + LangGraph            | Week 9-11  |
| FastAPI + PostgreSQL       | Week 11-12 |

## 6-Month Solid

| Topic                    | Timeline  |
|--------------------------|-----------|
| Python + DSA             | Month 1   |
| ML + Deep Learning       | Month 2   |
| NLP + Transformers       | Month 3   |
| LLM Engineering + RAG    | Month 3-4 |
| Fine-Tuning (LoRA/QLoRA) | Month 4   |
| Backend + DevOps         | Month 5   |
| Portfolio Projects x3    | Month 6   |

## 12-Month Expert

| Topic                           | Timeline  |
|---------------------------------|-----------|
| Python + DSA + ML Fundamentals  | Month 1-2 |
| Deep Learning + NLP             | Month 3-4 |
| LLM Engineering + RAG           | Month 4-5 |
| Fine-Tuning + Eval + Deployment | Month 5-6 |
| AI System Design                | Month 6-7 |
| Backend + FastAPI + PostgreSQL  | Month 7-8 |
| DevOps + Docker + CI/CD         | Month 8-9 |

|                              |             |
|------------------------------|-------------|
| 5 Portfolio Projects         | Month 9-11  |
| Interview Prep + Open Source | Month 11-12 |

## What To Skip (For This Internship)

|                                           |                                                        |
|-------------------------------------------|--------------------------------------------------------|
| ✗ PHP / Laravel / Livewire / Alpine.js    | Not needed — you're on the AI engineering track        |
| ✗ TensorFlow                              | Use PyTorch; TF is less dominant in research/LLM space |
| ✗ Bootstrap                               | Use Tailwind CSS instead                               |
| ✗ Advanced GraphQL                        | REST is sufficient for this stage                      |
| ✗ React Native / Mobile Dev               | Web-only for this project                              |
| ✗ Competitive Programming (hard LeetCode) | Easy-Medium is enough                                  |

## Interview Preparation

### ML Concepts to Know Cold

- Bias-Variance Tradeoff
- Overfitting & Regularisation (L1/L2)
- Gradient Descent variants (SGD, Adam, RMSprop)
- Cross-validation & K-Fold
- Precision vs Recall vs F1
- ROC-AUC interpretation
- Why Transformers replaced RNNs
- How Attention mechanism works

### NLP Interview Topics

- What is tokenization? Types?
- Explain BERT pre-training objectives
- BM25 vs dense retrieval
- How does RAG work end-to-end?
- What is hallucination? How do you reduce it?
- Cosine similarity in embedding space
- Explain LoRA in simple terms
- How do you evaluate an NLP model?

## System Design Questions

- Design a RAG pipeline for 1M documents
- How would you scale a multi-agent system?
- How do you handle latency in LLM APIs?
- Design a CI/CD pipeline for an ML model
- How do you monitor a deployed AI system?
- Explain the difference between synchronous and async APIs

## Your Identified Gaps (Fix These First)

- NLP fundamentals — practice explaining Transformers clearly
- Fine-tuning pipeline — walk through LoRA end-to-end
- System design — practice 1 AI design question daily
- ML math — review gradient descent derivation
- Evaluation metrics — know when to use what and why

# Master Skill Checklist

| Skill                                               | Area             | Done?                               |
|-----------------------------------------------------|------------------|-------------------------------------|
| ■ Python Advanced (OOP, async, type hints)          | Python Advanced  | <input checked="" type="checkbox"/> |
| ■ DSA: Arrays, Trees, Graphs, DP                    | DSA              | <input checked="" type="checkbox"/> |
| ■ ML: Regression, Classification, Clustering        | ML               | <input checked="" type="checkbox"/> |
| ■ Deep Learning: CNN, RNN, Transformers (PyTorch)   | Deep Learning    | <input checked="" type="checkbox"/> |
| ■ NLP: Tokenization, BERT, Embeddings, NER          | NLP              | <input checked="" type="checkbox"/> |
| ■ NLP: Semantic Search, Q&A, Text Classification    | NLP              | <input checked="" type="checkbox"/> |
| ■ LLM Engineering: Prompt Engineering, RAG          | LLM Engineering  | <input checked="" type="checkbox"/> |
| ■ LLM: LangChain, LangGraph, LlamaIndex             | LLM              | <input checked="" type="checkbox"/> |
| ■ Fine-Tuning: LoRA, QLoRA, SFT                     | Fine-Tuning      | <input checked="" type="checkbox"/> |
| ■ Fine-Tuning: Dataset prep, evaluation, deployment | Fine-Tuning      | <input checked="" type="checkbox"/> |
| ■ AI System Design: RAG arch, multi-agent           | AI System Design | <input checked="" type="checkbox"/> |
| ■ Backend: FastAPI, REST, JWT, WebSockets           | Backend          | <input checked="" type="checkbox"/> |
| ■ Database: PostgreSQL, SQLAlchemy, Alembic         | Database         | <input checked="" type="checkbox"/> |
| ■ DevOps: Docker, GitHub Actions, Nginx             | DevOps           | <input checked="" type="checkbox"/> |
| ■ Frontend: React, Next.js, Tailwind CSS            | Frontend         | <input checked="" type="checkbox"/> |
| ■ Portfolio: 3–5 deployed AI projects               | Portfolio        | <input checked="" type="checkbox"/> |